Regression Analysis in Excel – Cheat Sheet

By: Shantanu Dhara | Day 14 — Data Analysis Skill Improvement Challenge

❖ What is Regression Analysis?

- Regression Analysis is a statistical method used to **find relationships between variables** — it helps us understand **how one variable changes when another changes** and predict future outcomes.

❖ Dataset Example:

✓ We are using a **Retail Store dataset** with the following columns:
Sales | Quantity | Discount | Profit

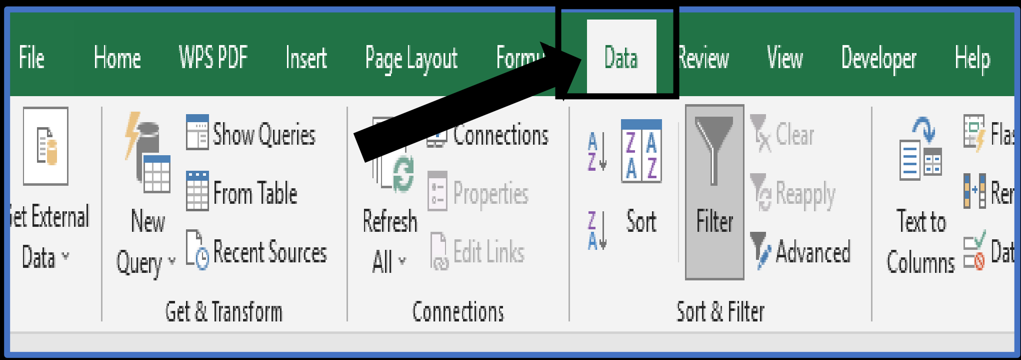
✓ Goal: To find out which factors (*Sales, Quantity, or Discount*) affect **Profit** the most.

Q	R	S	T
sales	quantity	discount	profit
262	2	0	42
732	3	0	220
15	2	0	7
958	5	0	-383
22	2	0	3
49	7	0	14
7	4	0	2
907	6	0	91
19	3	0	6
115	5	0	34
1706	9	0	85
911	4	0	68
16	3	0	5
408	3	0	133
69	5	1	-124
3	3	1	-4
666	6	0	13
56	2	0	10
9	2	0	2
213	3	0	16
23	4	0	7
[illegible]	[illegible]	[illegible]	[illegible]

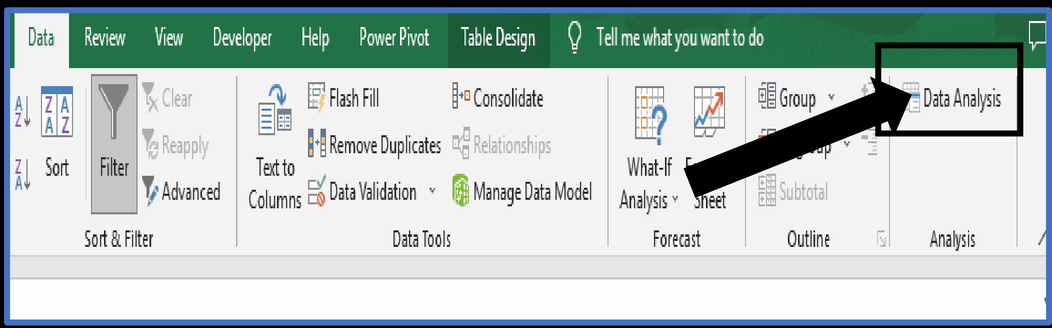
❖ Steps to Create a Regression Model in Excel

✓ Step 1: Open the Data Analysis Tool

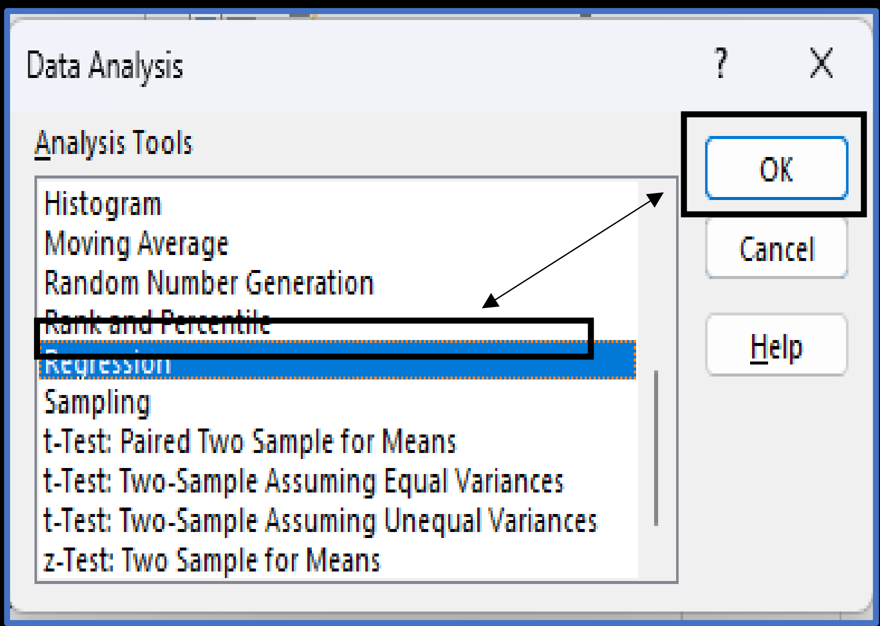
- Go to the **Data tab** on the ribbon.



- Click **Data Analysis** (under the *Analysis* group).

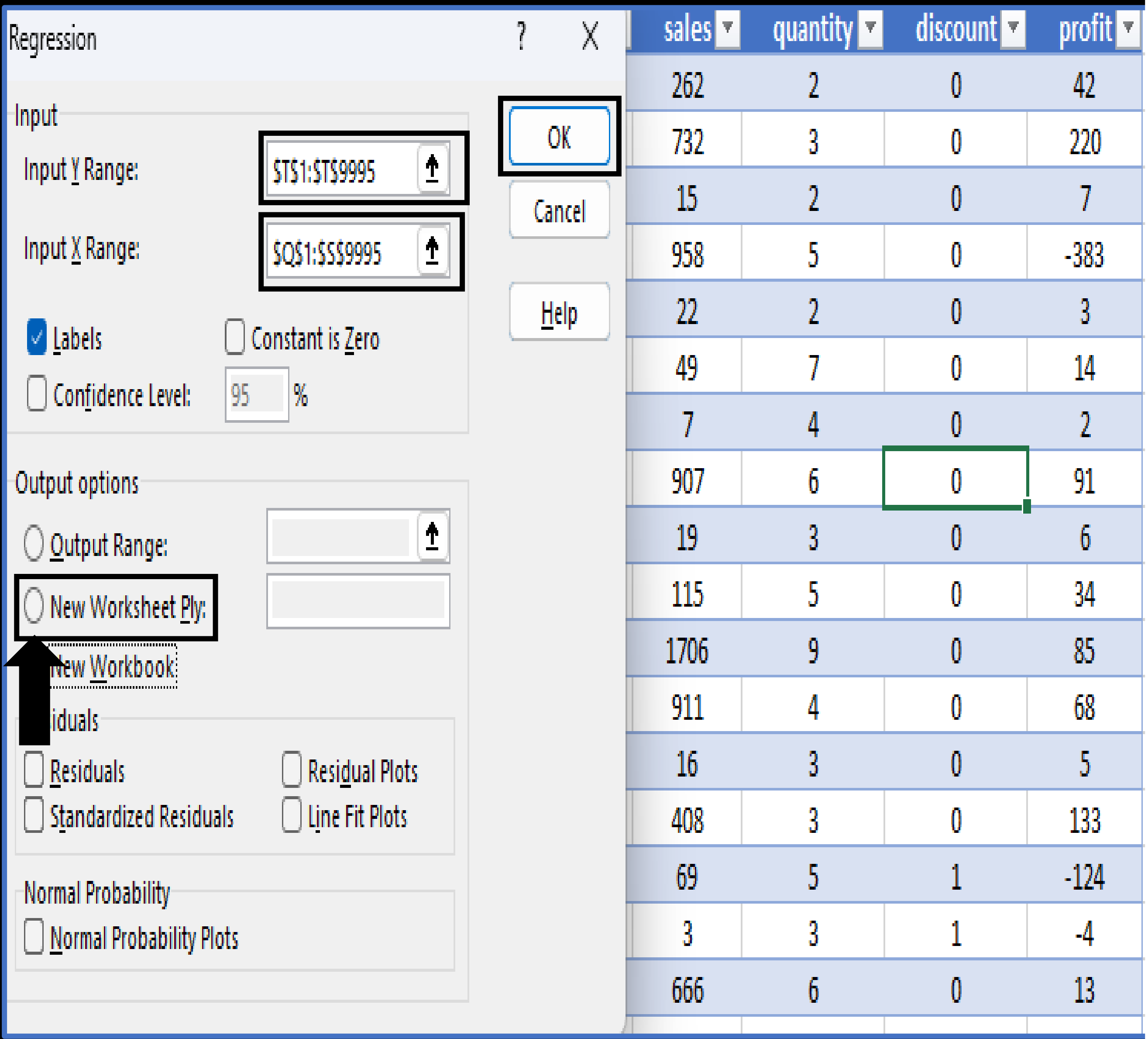


- A popup window will appear and From the list, select **Regression** → click **OK**.



✓ Step 2: Input Range Selection

- **Input Y Range (Dependent Variable):**
  → Select the **Profit** column (this is the outcome we're predicting).
- **Input X Range (Independent Variables):**
  → Select **Sales**, **Quantity**, and **Discount** columns (these are the predictors).
- **Labels:**
  → Check this box if your first row has column names.
- Choose **New Worksheet Ply** (recommended for clear results).
- Leave **Constant is Zero** and **Confidence Level** as default.
- Click **OK** to generate results.



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✓

✓ Step 3: Understanding the Regression Output

Term	**Meaning**
R Square (R^2)	Explains how well independent variables predict profit. Higher = better fit.
Coefficients	Show direction and size of impact (positive or negative).
P-Value	Checks significance of each factor. $P < 0.05$ = significant impact.
Intercept	Base value of profit when other variables are zero.

Once you click OK, Excel gives a detailed output. Here's how to interpret it:

SUMMARY OUTPUT

Regression Statistics	
Multiple R	0.522230578
R Square	0.272724777
Adjusted R Square	0.272506376
Standard Error	199.8080283
Observations	9994

ANOVA

	df	*SS*	*MS*	*F*	*Significance F*
Regression	3	149560586.6	49853528.86	1248.73428	0
Residual	9990	398833249.4	39923.24819		
Total	9993	548393836			

	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*	*Lower 95.0%*	*Upper 95.0%*
Intercept	34.97212971	4.218054423	8.291057014	1.26699E-16	26.7038932	43.24036622	26.7038932	43.24036622
sales	0.180010144	0.003275256	54.96063281	0	0.173589982	0.186430306	0.173589982	0.186430306
quantity	-2.962176487	0.917056761	-3.230090668	0.001241514	-4.759792505	-1.164560468	-4.759792505	-1.164560468
discount	-233.4570011	9.686453543	-24.10139068	8.4427E-125	-252.4444017	-214.4696006	-252.4444017	-214.4696006

Thanks For Watching | Shantanu Dhara